



Guangdong Technion
Israel Institute of Technology
广东以色列理工学院

035033-Intro to Sensor-Integrated Systems

Lecture II: Accuracy of Measurement Systems in the Steady State

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Rules for the course

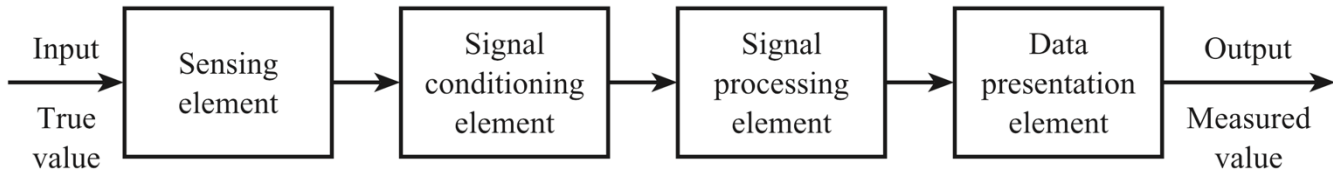
- ❑ **Homework** have to have the final answer in a **red** box.
- ❑ The answer must include a unit. If not, it is wrong. 5 Ampere is right, but 5 is wrong.
- ❑ Copy homework from others is strictly forbidden.
- ❑ Sharing homework to others is also not allowed, the same punishment.
- ❑ Legal way: come and take a look use eyes -- no photo, no copy, no Wechat, no email sent
- ❑ Homework needs to be very well organized.
- ❑ Welcome to discuss, check final answer.
- ❑ For homework: 90% of students gets 100% correct.
- ❑ It does not make sense to copy from others, as it only takes 20% of all the credits.
- ❑ There is no way to copy during the exam, which accounts 60% of the final score.

5 A ✓

5 ✗

Structure of measurement systems

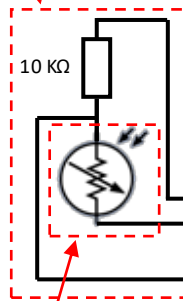
- ❑ **Sensing element** is in contact with the process and gives an output which depends in some way on the variable to be measured.
- ❑ **Signal conditioning element** takes the output of the sensing element and converts it into a form more suitable for further processing, usually a D.C. voltage/current or frequency signal.
- ❑ **Signal processing element** takes the output of the conditioning element and converts it into a form more suitable for presentation. Example: Analogue-to-digital converter (ADC)
- ❑ **Data presentation element** presents the measured value in a form which can be easily recognized by the observer.



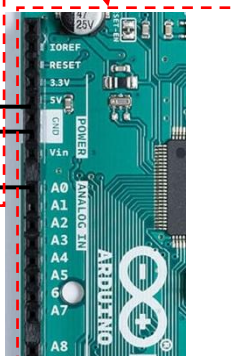
Structure of measurement systems



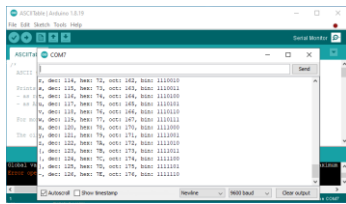
Signal conditioning element: Voltage divider



Signal processing element: ADC



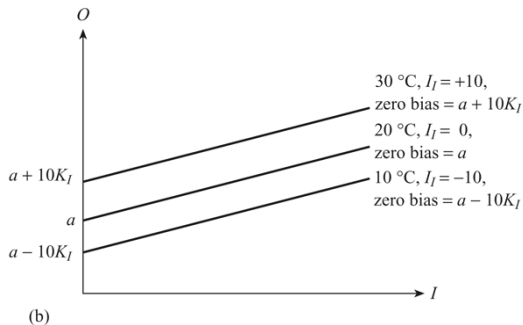
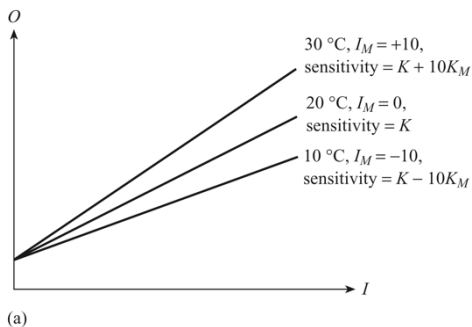
Sensing element:
Photoresistor



← Data presentation element: Serial monitor

Environmental effects

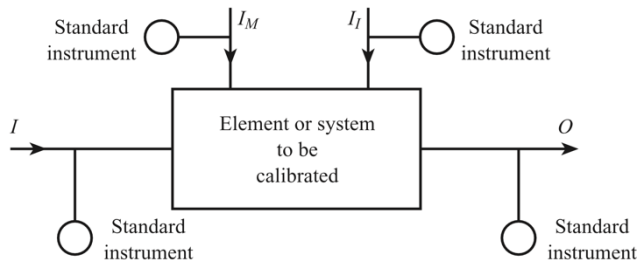
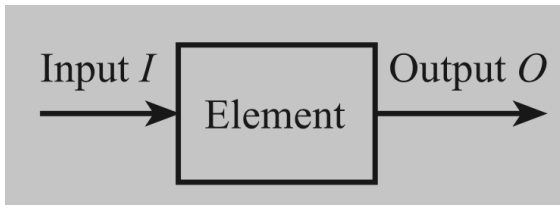
- ❑ $O = KI + a + N(I) + K_M I_M I + K_I I_I = (K + K_M I_M)I + (a + K_I I_I) + N(I)$
- ❑ I_M : modifying input, is an environmental variable that changes the sensitivity (slope) of the sensor. K_M is the change in sensitivity for unit change in I_M causes the linear sensitivity of an element to change.
- ❑ I_I : Interfering input, an environmental variable that causes a change in the zero offset (intercept) of the sensor output. K_I : is the change in sensitivity for unit change in I_I causes the zero offset to change.



Identification of static characteristics – calibration

❑ $O = KI + a + N(I) + K_M I_M I + K_I I_I$

- ❑ The static characteristics of an element can be found experimentally by measuring corresponding values of the input I , the output O and the environmental inputs I_M and I_I , when I is either at a constant value or changing slowly. This type of experiment is referred to as calibration, and the measurement of the variables I , O , I_M and I_I must be accurate if meaningful results are to be obtained. The instruments and techniques used to quantify these variables are referred to as standards.



Hint for homework

2.7 The following results were obtained when a pressure transducer was tested in a laboratory under the following conditions:

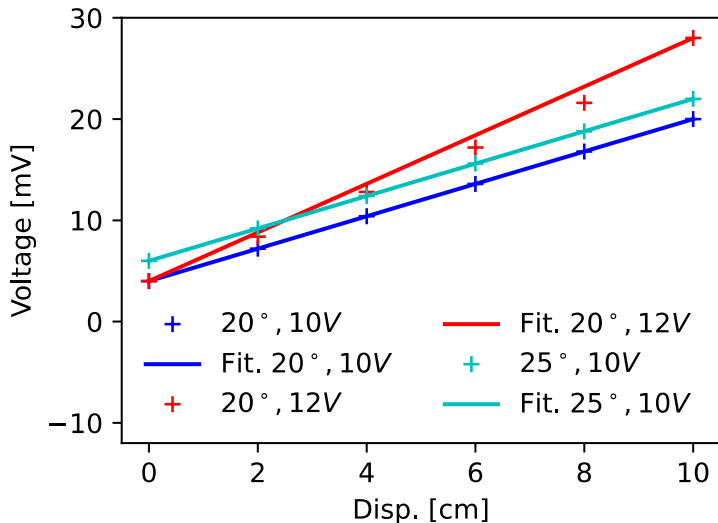
- I Ambient temperature 20 °C, supply voltage 10 V (standard)
- II Ambient temperature 20 °C, supply voltage 12 V
- III Ambient temperature 25 °C, supply voltage 10 V

Input (barg)	0	2	4	6	8	10
Output (mA)						
I	4	7.2	10.4	13.6	16.8	20
II	4	8.4	12.8	17.2	21.6	28
III	6	9.2	12.4	15.6	18.8	22

- (a) Determine the values of K_M , K_I , a and K associated with the generalised model equation $O = (K + K_M I_M)I + a + K_I I_I$.
- (b) Predict an output value when the input is 5 barg, $V_S = 12$ V and ambient temperature is 25 °C.

$O = (K + K_M I_M)I + a + N(I) + K_I I_I$

All come from standard sensors.



Lecture 2

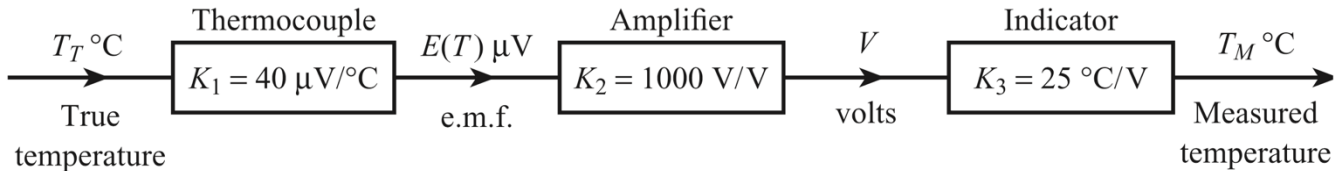
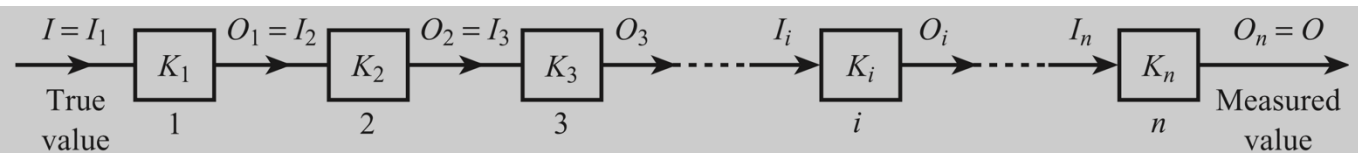
❑ Preview study: *Principles of Measurement Systems*, Chapter 3: The Accuracy of Measurement Systems in the Steady State

An ideal displacement sensor can be described by $V = KI + a = 0.012I + 0.2$, at 20°C and with power supply voltage 5 V. Where V is voltage output in *Volts*, input I is displacement in meters, and a is bias in *Volts*. However, the power supply voltage (interfering input I_I) and the environment temperature (modifying input I_M) will affect the output. The general model that describes the behavior is $O = KI + a + N(I) + K_M I_M I + K_I I_I$. The nonlinear effect $N(I)$ can be ignored. The coefficients are $K_M = 0.0002 \frac{\text{V}}{\text{m}^\circ\text{C}}$ and $K_I = 0.01 \frac{\text{V}}{\text{V}}$.

- a) What is the sensitivity and bias when the displacement sensor is in 30°C environment and with power supply voltage 5.5 V.
- b) What is the output with the idea model when measuring 40 cm displacement and the model that considers environment effect in a)?
- c) How much is the error in percentage if the environment factor is not considered?

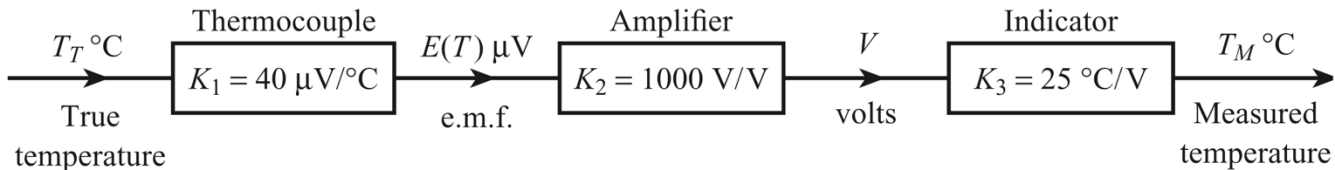
Measurement error of a system of ideal elements

- ❑ $E = O - I = IK_1K_2 \dots K_n - I = (K_1K_2 \dots K_n - 1)I$
- ❑ If $K_1K_2 \dots K_n = 1$, $E = 0$, perfectly accurate sensor
- ❑ However, non-ideal characteristics in every element, *e.g.*, non-linearity, environmental and statistical effects



Measurement Error Exercise

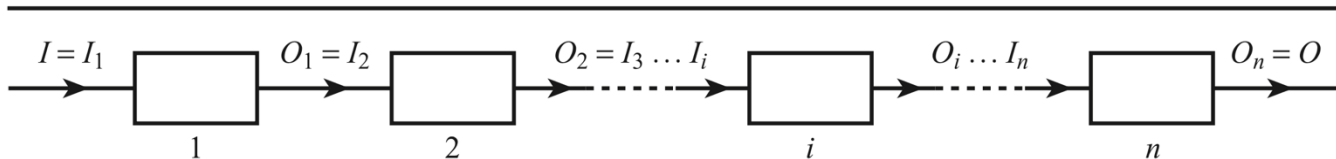
□ How much is the error if $K_1 = 38 \mu\text{V}/^\circ\text{C}$, and $T_T = 20^\circ\text{C}$



Error probability density function

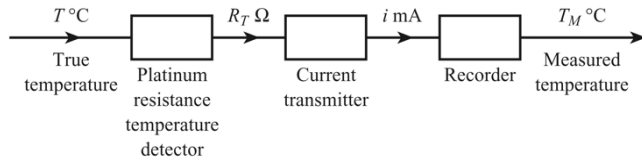
- ❑ $O = KI + a + N(I) + K_M I_M I + K_I I_I$
- ❑ $\bar{O}_1 = \bar{K}_1 \bar{I}_1 + \bar{a}_1 + \bar{N}_1(\bar{I}_1) + \bar{K}_{M1} \bar{I}_{M1} \bar{I}_1 + \bar{K}_{I1} \bar{I}_{I1}$
- ❑ $\bar{O}_2 = \bar{K}_2 \bar{I}_2 + \bar{a}_2 + \bar{N}_2(\bar{I}_2) + \bar{K}_{M2} \bar{I}_{M2} \bar{I}_2 + \bar{K}_{I2} \bar{I}_{I2}$
- ❑ ...
- ❑ $\bar{O} = \bar{O}_n = \bar{K}_n \bar{I}_n + \bar{a}_n + \bar{N}_n(\bar{I}_n) + \bar{K}_{Mn} \bar{I}_{Mn} \bar{I}_n + \bar{K}_{In} \bar{I}_{In}$
- ❑ Error $\bar{E} = \bar{O} - \bar{I}$ probability density function

$$p(E) = \frac{1}{\sigma_E \sqrt{2\pi}} \exp \left[-\frac{1}{2\sigma_E^2} (E - \bar{E})^2 \right]$$



$$\sigma_{I_2}^2 = \sigma_{O_1}^2 = \left(\frac{\partial O_1}{\partial I_1} \right)^2 \sigma_{I_1}^2 + \left(\frac{\partial O_1}{\partial I_{M1}} \right)^2 \sigma_{I_{M1}}^2 + \left(\frac{\partial O_1}{\partial I_{I1}} \right)^2 \sigma_{I_{I1}}^2 + \left(\frac{\partial O_1}{\partial K_1} \right)^2 \sigma_{K_1}^2 + \dots$$

Error probability density function



(a) Platinum resistance temperature detector

Model equation

$$R_T = R_0(1 + \alpha T + \beta T^2)$$

Individual mean values

$$\bar{R}_0 = 100.0 \, \Omega, \bar{\alpha} = 3.909 \times 10^{-3}, \bar{\beta} = -5.897 \times 10^{-7} \text{ (between 100 and 130 °C)}$$

Individual standard deviations

$$\sigma_{R_0} = 4.33 \times 10^{-2}, \sigma_{\alpha} = 0.0, \sigma_{\beta} = 0.0$$

Partial derivatives

$$\frac{\partial R_T}{\partial R_0} = 1.449 \text{ at } T = 117 \, ^\circ\text{C}$$

Overall mean value

$$\bar{R}_T = \bar{R}_0(1 + \bar{\alpha}\bar{T} + \bar{\beta}\bar{T}^2)$$

Overall standard deviation

$$\sigma_{R_T}^2 = \left(\frac{\partial R_T}{\partial R_0} \right)^2 \sigma_{R_0}^2$$

$$\bar{R}_T$$

$$= \bar{R}_0(1 + \bar{\alpha}\bar{T} + \bar{\beta}\bar{T}^2)$$

$$= 100$$

$$* (1 + 3.909 * 10^{-3}$$

$$* 117 - 5.897 * 10^{-7}$$

$$* 117^2) = 144.9 \, \Omega$$

$$\frac{\partial R_T}{\partial R_0} = 1 + \alpha T + \beta T^2$$

$$= 1 + \bar{\alpha}\bar{T} + \bar{\beta}\bar{T}^2$$

$$= 1 + 3.909 * 10^{-3}$$

$$* 117 - 5.897 * 10^{-7}$$

$$* 117^2 = 1.449$$

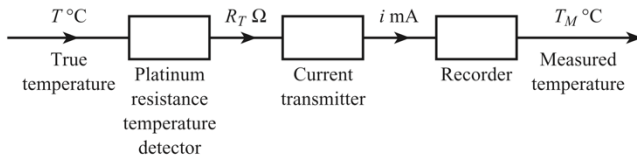
$$\sigma_{R_T}^2 = \left(\frac{\partial R_T}{\partial R_0} \right)^2 \sigma_{R_0}^2$$

$$= (1.449)^2$$

$$* (4.33 * 10^{-2})^2$$

$$= 0.00394$$

Error probability density function



(b) Current transmitter

Model equation 4 to 20 mA output for 138.5 to 149.8 Ω input (100 to 130 °C)

ΔT_a = deviation of ambient temperature from 20 °C

$$i = KR_T + K_M R_T \Delta T_a + K_I \Delta T_a + a$$

Individual mean values

$$\bar{K} = 1.4134, \bar{K}_M = 1.4134 \times 10^{-4}, \bar{K}_I = -1.637 \times 10^{-2}$$

$$\bar{a} = -191.76, \Delta \bar{T}_a = -10$$

Individual standard deviations

$$\sigma_K = 0.0, \sigma_{K_M} = 0.0, \sigma_{K_I} = 0.0$$

$$\sigma_a = 0.24, \sigma_{\Delta T_a} = 6.7$$

Partial derivatives

$$\frac{\partial i}{\partial R_T} = 1.413, \frac{\partial i}{\partial \Delta T_a} = 4.11 \times 10^{-3}, \frac{\partial i}{\partial a} = 1.00$$

Overall mean value

$$\bar{i} = \bar{K} \bar{R}_T + \bar{K}_M \bar{R}_T \Delta \bar{T}_a + \bar{K}_I \Delta \bar{T}_a + \bar{a}$$

Overall standard deviation

$$\sigma_i^2 = \left(\frac{\partial i}{\partial R_T} \right)^2 \sigma_{R_T}^2 + \left(\frac{\partial i}{\partial \Delta T_a} \right)^2 \sigma_{\Delta T_a}^2 + \left(\frac{\partial i}{\partial a} \right)^2 \sigma_a^2$$

(c) Recorder

Model equation

$$T_M = Ki + a$$

Individual mean values

$$\bar{K} = 1.875, \bar{a} = 92.50$$

(100 to 130 °C record for 4 to 20 mA input)

Individual standard deviations

$$\sigma_K = 0.0, \sigma_a = 0.10$$

Partial derivatives

$$\frac{\partial T_M}{\partial i} = 1.875, \frac{\partial T_M}{\partial a} = 1.00$$

Overall mean value

$$\bar{T}_M = \bar{K} \bar{i} + \bar{a}$$

Overall standard deviation

$$\sigma_{T_M}^2 = \left(\frac{\partial T_M}{\partial i} \right)^2 \sigma_i^2 + \left(\frac{\partial T_M}{\partial a} \right)^2 \sigma_a^2$$

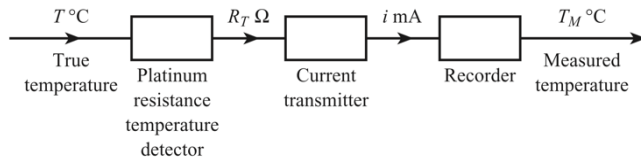
Error probability density function

Mean $\bar{E}$

$$\bar{T} = 117\text{ }^{\circ}\text{C} \quad \bar{R}_T = 144.93\text{ }\Omega$$

$$\bar{i} = 13.04\text{ mA} \quad \bar{T}_M = 116.95\text{ }^{\circ}\text{C}$$

$$\bar{E} = \bar{T}_M - \bar{T} = -0.005\text{ }^{\circ}\text{C}$$



Standard deviation σ_E

$$\sigma_T^2 = 0$$

$$\sigma_{R_T}^2 = \left(\frac{\partial R_T}{\partial R_0} \right)^2 \sigma_{R_0}^2 = 39.4 \times 10^{-4}$$

$$\begin{aligned} \sigma_i^2 &= \left(\frac{\partial i}{\partial R_T} \right)^2 \sigma_{R_T}^2 + \left(\frac{\partial i}{\partial \Delta T_a} \right)^2 \sigma_{\Delta T_a}^2 + \left(\frac{\partial i}{\partial a} \right)^2 \sigma_a^2 \\ &= 78.7 \times 10^{-4} + 8.18 \times 10^{-4} + 5.76 \times 10^{-2} \\ &= 6.62 \times 10^{-2} \end{aligned}$$

$$\sigma_{T_M}^2 = \left(\frac{\partial T_M}{\partial i} \right)^2 \sigma_i^2 + \left(\frac{\partial T_M}{\partial a} \right)^2 \sigma_a^2 = 24.3 \times 10^{-2}$$

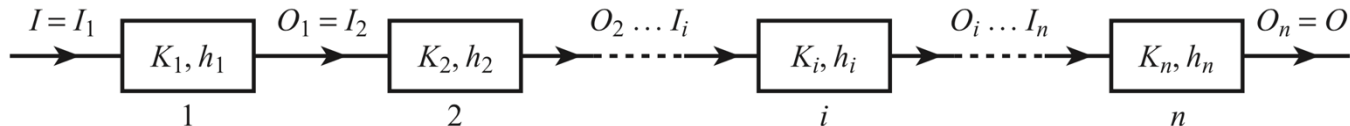
$$\sigma_E = \sigma_{T_M} = 0.49\text{ }^{\circ}\text{C}$$

Exercise

A measurement system consists of a chromel–alumel thermocouple (with cold junction compensation), a millivolt-to-current converter and a recorder. Table Prob. 1 gives the model equations, and parameters for each element. Assuming that all probability distributions are normal, calculate the mean and standard deviation of the error probability distribution, when the input temperature is 117 °C.

	Chromel–alumel thermocouple	e.m.f-to-current converter	Recorder
Model equation	$E = C_0 + C_1T + C_2T^2$	$i = K_1E + K_M E \Delta T_a + K_I \Delta T_a + a_1$	$T_M = K_2i + a_2$
Mean values	$\bar{C}_0 = 0.00$ $\bar{C}_1 = 4.017 \times 10^{-2}$ $\bar{C}_2 = 4.66 \times 10^{-6}$	$\bar{K}_1 = 3.893$ $\Delta T_a = -10$ $\bar{a}_1 = -3.864$ $\bar{K}_M = 1.95 \times 10^{-4}$ $\bar{K}_I = 2.00 \times 10^{-3}$	$\bar{K}_2 = 6.25$ $\bar{a}_2 = 25.0$
Standard deviations	$\sigma_{C_0} = 6.93 \times 10^{-2}$ $\sigma_{C_1} = \sigma_{C_2} = 0$	$\sigma_{a_1} = 0.14, \sigma_{\Delta T_a} = 10$ $\sigma_{K_1} = \sigma_{K_M} = \sigma_{K_I} = 0$	$\sigma_{a_2} = 0.30$ $\sigma_{K_2} = 0.0$

Modelling using error bands



$$\bar{O}_i = K_i \bar{I}_i$$

$$\bar{O} = K_1 K_2 \dots K_i \dots K_n \bar{I}$$

$$\bar{E} = \bar{O} - \bar{I} \text{ [} = 0, \text{ if } K_1 K_2 \dots K_n = 1 \text{]}$$

$$\sigma_{I_1}^2 = 0$$

$$\sigma_{I_2}^2 = \sigma_{O_1}^2 = \frac{h_1^2}{3}$$

$$\sigma_{I_3}^2 = \sigma_{O_2}^2 = K_2^2 \sigma_{I_2}^2 + \frac{h_2^2}{3}$$

$$\vdots \quad \vdots$$

$$\sigma_{I_{i+1}}^2 = \sigma_{O_i}^2 = K_i^2 \sigma_{I_i}^2 + \frac{h_i^2}{3}$$

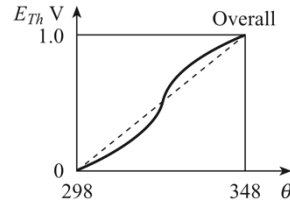
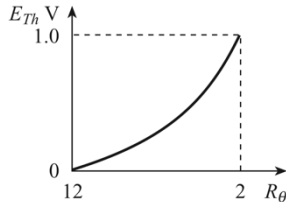
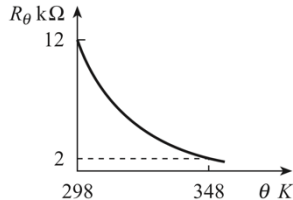
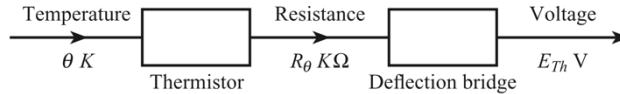
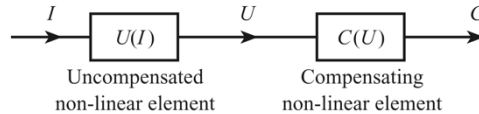
$$\vdots \quad \vdots$$

$$\sigma_O^2 = \sigma_{O_n}^2 = K_n^2 \sigma_{I_n}^2 + \frac{h_n^2}{3}$$

$$\sigma_E = \sigma_0$$

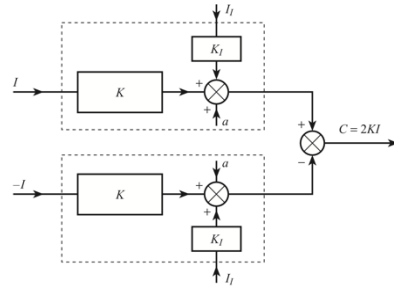
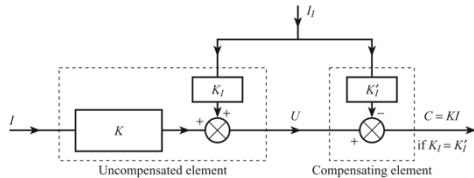
Error reduction techniques

- ❑ Correcting a non-linear element is to introduce a compensating non-linear element



Reducing the effects of environment inputs

- ❑ The most obvious method of reducing the effects of environmental inputs is that of isolation, i.e., to isolate the transducer from environmental changes so that effectively $I_M = I_I = 0$.
- ❑ Examples of this are the placing of the reference junction of a thermocouple in a temperature-controlled enclosure, and the use of spring mountings to isolate a transducer from the vibrations of the structure to which it is attached.



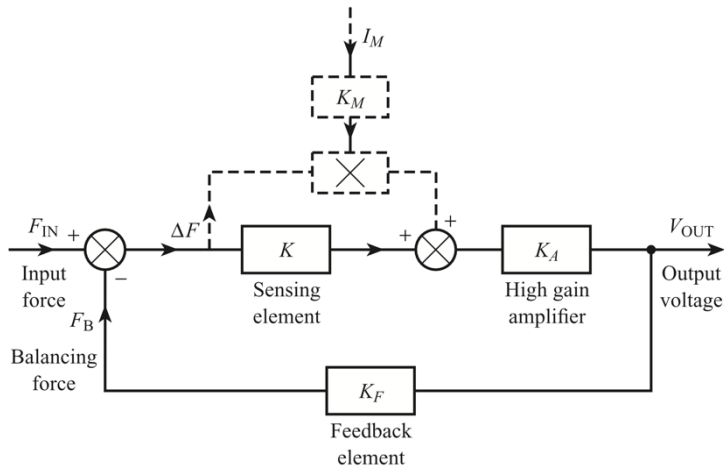
Reducing the effects of environment inputs

- ❑ Another obvious method is that of zero environmental sensitivity, where the element is completely insensitive to environmental inputs, *i.e.*, $K_M = K_I = 0$.
- ❑ An example of this is the use of a metal alloy with zero temperature coefficients of expansion and resistance as a strain gauge element. Such an ideal material is difficult to find, and in practice the resistance of a metal strain gauge is affected slightly by changes in ambient temperature.
- ❑ A more successful method of coping with environmental inputs is that of opposing environmental inputs. Suppose that an element is affected by an environmental

High-gain feedback

- The use of high-gain negative feedback is an important method of compensating for modifying inputs and non-linearity

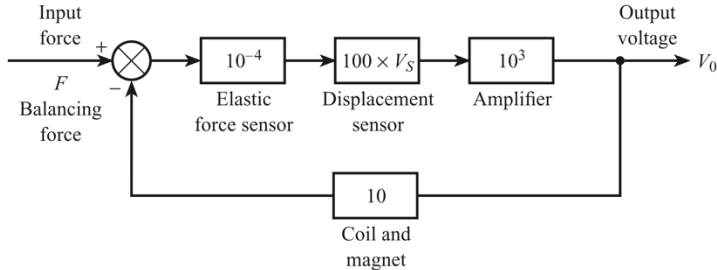
$$V_{\text{OUT}} = \frac{KK_A}{1 + KK_A K_F} F_{\text{IN}}$$



3.3

Figure Prob. 3 shows a block diagram of a force transducer using negative feedback. The elastic sensor gives a displacement output for a force input; the displacement sensor gives a voltage output for a displacement input. V_s is the supply voltage for the displacement sensor.

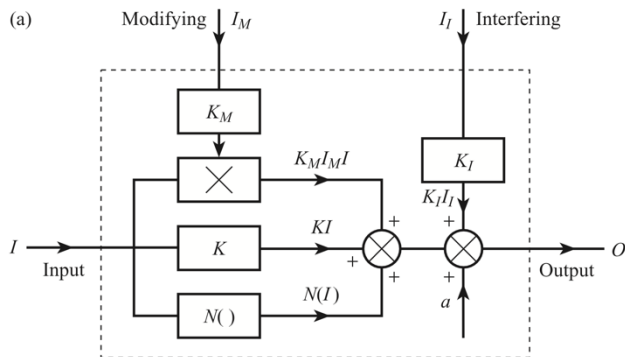
- (a) Calculate the output voltage V_0 when
- (i) $V_s = 1.0 \text{ V}$, $F = 50 \text{ N}$
 - (ii) $V_s = 1.5 \text{ V}$, $F = 50 \text{ N}$.
- (b) Comment on the practical significance of the variation of the supply voltage V_s .



Direct equation vs. Inverse equation

❑ Direct $O = KI + a + N(I) + K_M I_M I + K_I I_I$

❑ Inverse $I = K'O + a' + N'(O) + K'_M I_M O + K'_I I_I$

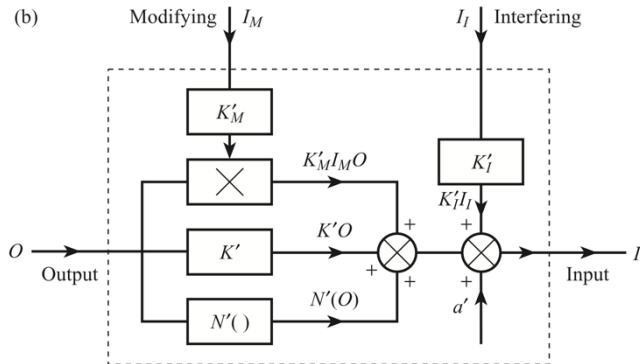


DIRECT

$$E = 3.845 \times 10^{-2} T + 4.682 \times 10^{-5} T^2 - 3.789 \times 10^{-8} T^3 + 1.652 \times 10^{-11} T^4 \text{ mV}$$

INVERSE

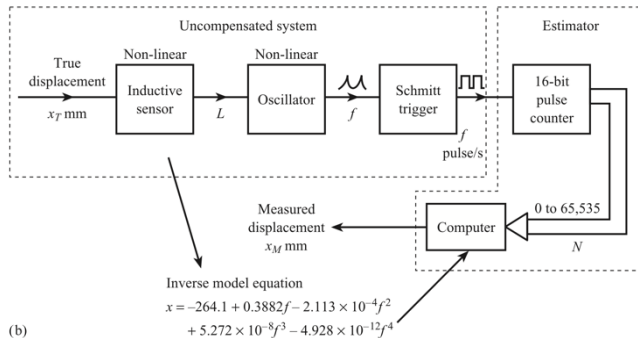
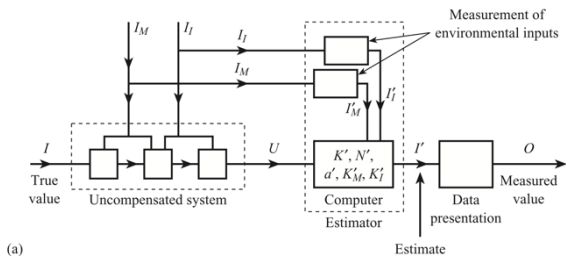
$$T = 25.55 E - 0.5973 E^2 + 2.064 \times 10^{-2} E^3 - 3.205 \times 10^{-4} E^4 \text{ } ^\circ\text{C}$$



[3.18]

Compensation

- ❑ **Compensation** is the deliberate introduction of a counteracting or modifying effect into a measurement system to reduce or eliminate an undesired influence



3.6

A fluid velocity measurement system consists of a pitot tube, a differential pressure transmitter, an 8-bit analogue-to-digital converter and a microcontroller with display facilities. Table Prob. 6 gives the model equations and parameters for each element in the system. The microcontroller calculates the measured value of velocity assuming a constant density.

- (a) Estimate the mean and standard deviation of the error probability density function assuming the true value of velocity v_T is 14.0 m s^{-1} . Use the procedure of Table 3.1, treating the rectangular distributions as normal with $\sigma = h/\sqrt{3}$.
- (b) Explain briefly what modifications could be made to the system to reduce the quantities calculated in (a).

	Pitot tube	Differential pressure transmitter	Analogue-to-digital converter	Microcontroller with display
Model equations	$\Delta P = \frac{1}{2}\rho v_T^2$	$i = K_1\Delta P + a_1$	$n = K_2i + a_2$	$v_M = K_3\sqrt{n - 51}$
Mean values	$\bar{\rho} = 1.2$	$\bar{K}_1 = 0.064$ $\bar{a}_1 = 4.0$	$\bar{K}_2 = 12.80$ $\bar{a}_2 = 0.0$ n rounded off to nearest integer	$\bar{K}_3 = 1.430$
Half-widths of rectangular distribution	$h_\rho = 0.1$	$h_{a_1} = 0.04$	$h_{a_2} = 0.5$	$h_{K_3} = 0.0$